

IEORE4004

Optimization Models and Methods

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Outline

- Minimum Spanning Tree Problem
 - LP Form
 - Prim's Algorithm
 - Kruskal's Algorithm
 - Example | The Offshore Wind Farming Problem

Example: Project Management Problem

Background:

A company wants to introduce a new model of its ever-popular cell phone to the market. To gauge the demand, a limited batch of product is to be brought to the market. The product is assembled from two parts, a circuit board, and housing. In addition, workers need to be trained and raw material procured.

Example: Project Management Problem

After the initial assessment, the project manager put together a list of all tasks along with estimated duration of each task. The tasks are interdependent, some need to finish before others can start.

First, (A) workers need to be trained and (B) raw materials purchased which takes 6 and 9 days, respectively. After these tasks are finished, both parts of the product, (C) the circuit board and (D) the housing, are manufactured taking 7 and 8 days, respectively. Each circuit board, once manufactured, needs to undergo additional testing (E) to comply with FCC regulation which takes 10 days. Afterwards, the cell phone is assembled, packaged, and shipped (F) which takes 12 days.

Example: Project Management Problem

Question:

What is the minimum number of days before the product can reach the market? What is the critical path of the production, i.e., the sequence of tasks delaying any of which delays the whole production?

Example: Project Management Problem

We identify main checkpoints:

- (1) start, (2) completion of tasks A and B,
- (3) completion of task D, (4) completion of tasks C, E, and
- (5) finish

Nodes {1, 2, 3, 4, 5}

Example: Project Management Problem

We identify main checkpoints:

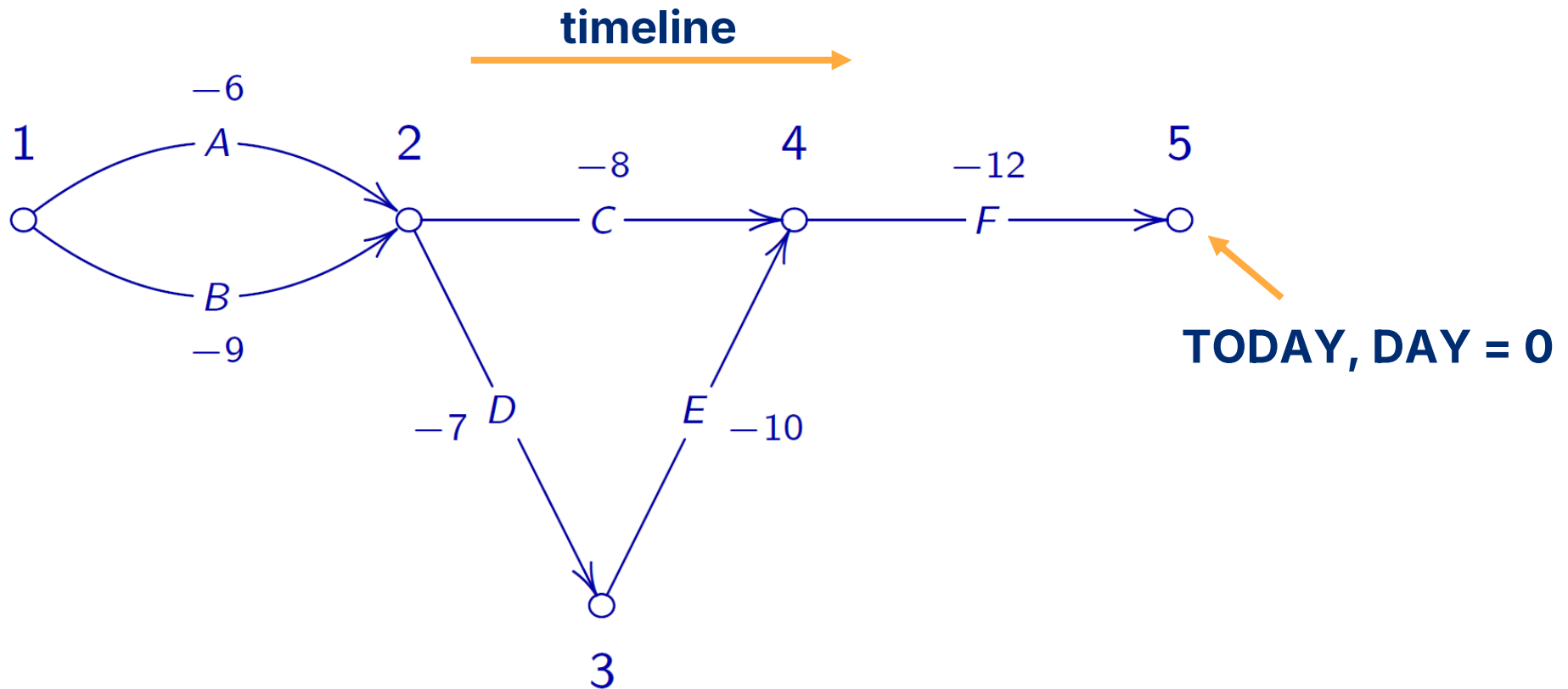
- (1) start, (2) completion of tasks A and B,
- (3) completion of task D, (4) completion of tasks C, E, and
- (5) finish

Nodes {1, 2, 3, 4, 5}

- ! Edges correspond to tasks that connect checkpoints, weights are durations with negative sign.

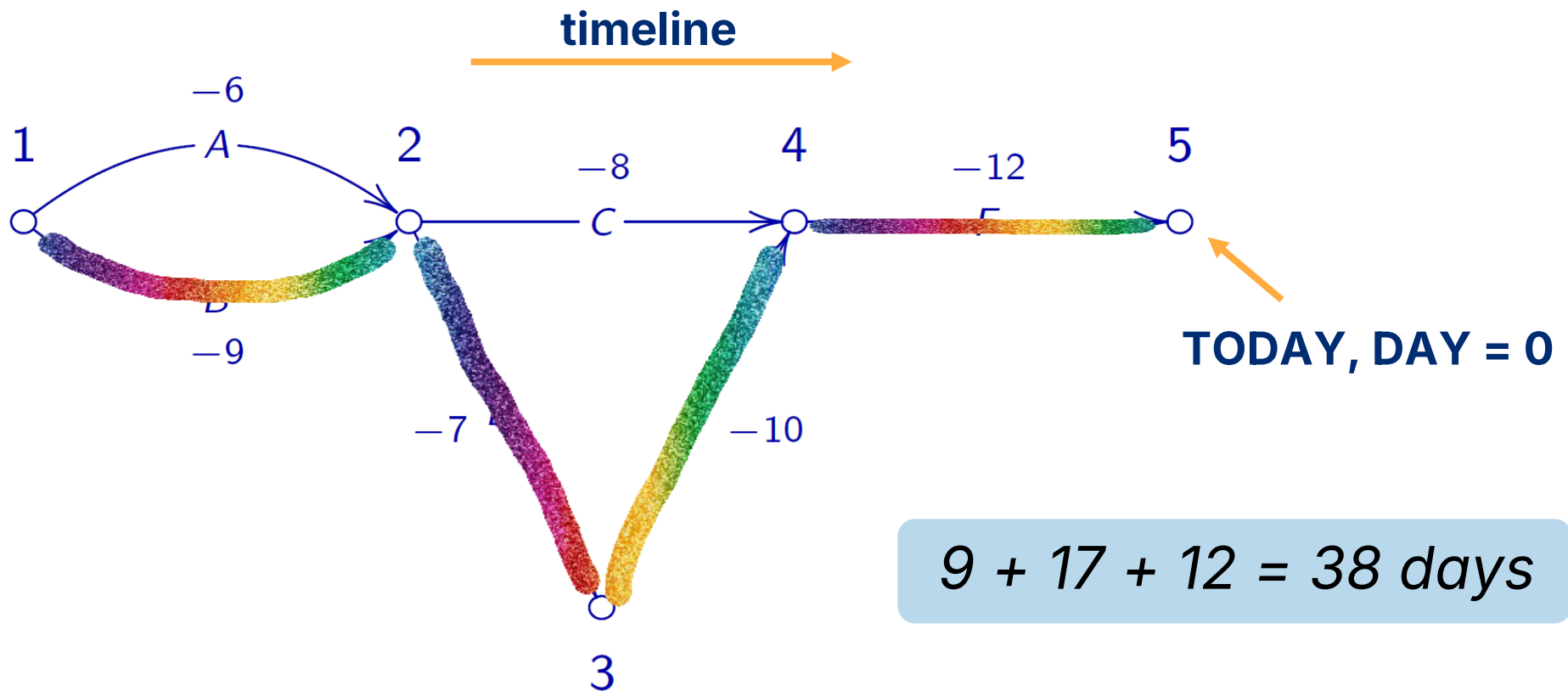
Example: Project Management Problem

Represent this in the network form



Example: Project Management Problem

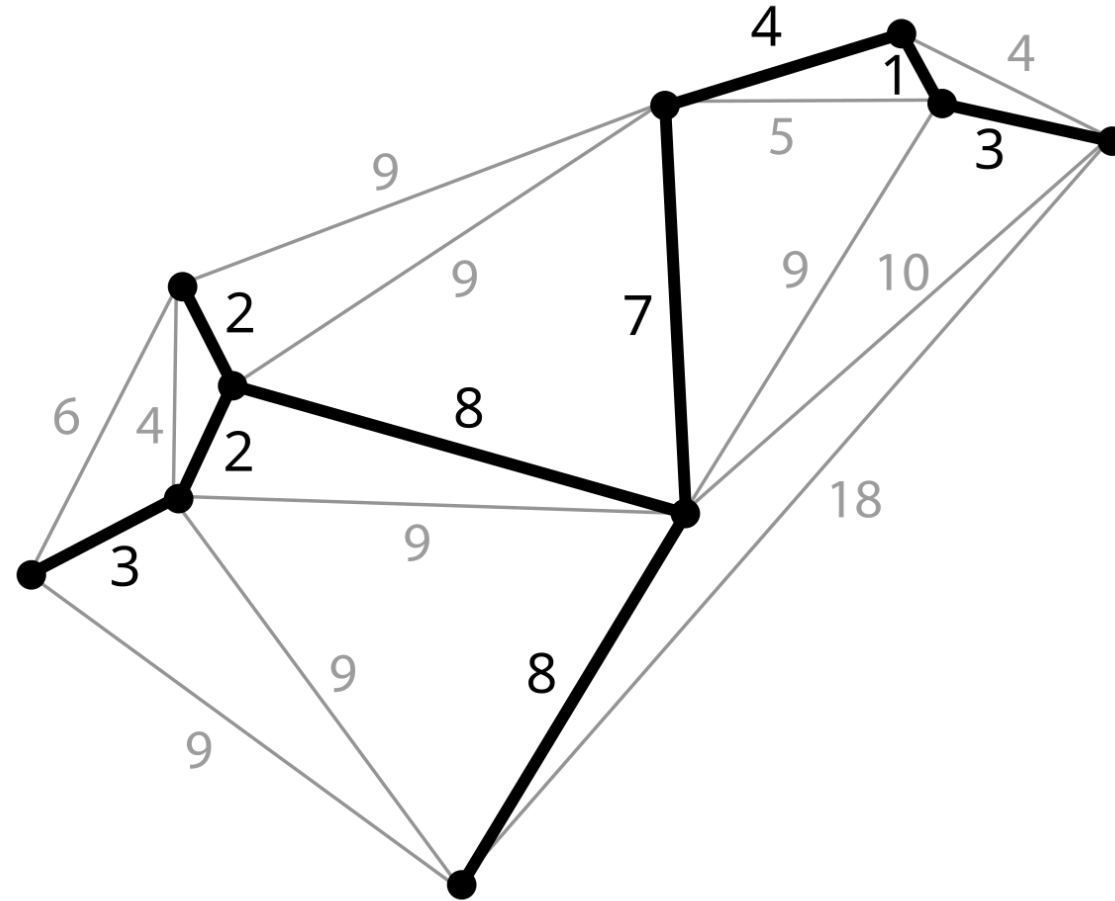
Represent this in the network form



Minimum Spanning Tree Problem

- LP Relaxation
- Kruskal's Algorithm

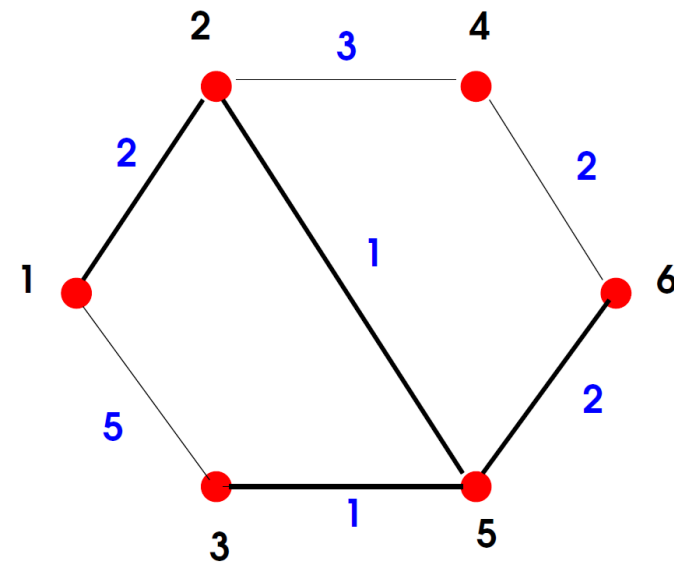
Minimum Spanning Tree Problem



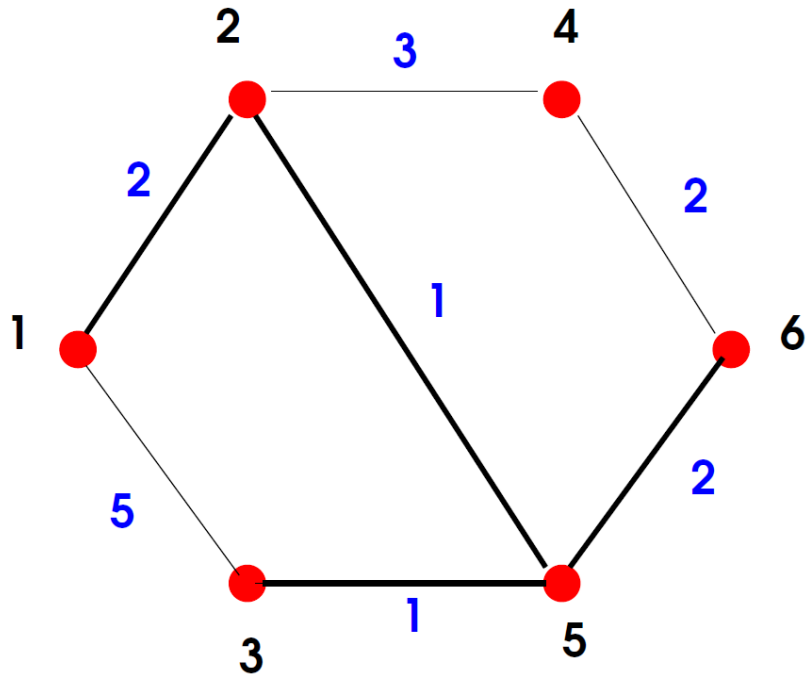
Minimum Spanning Tree

- For a network with n nodes, a **spanning tree** is a group of $n-1$ arcs that connects all nodes of the network and contains no loops.
- A spanning tree of minimum length in a network is a **minimum spanning tree (MST)**

Not spanning because it does not contain the vertex 4



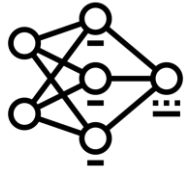
Minimum Spanning Tree



How can we find the minimum spanning tree and complete this network?

- Adding the edge **(2, 4)** yields a spanning tree of weight 9
- Adding the edge **(4, 6)** yields a spanning tree of weight 8
- Adding the edge **(1, 3)** is not possible as it creates a cycle

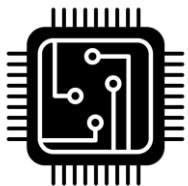
Minimum Spanning Tree Use Cases



Telecommunication Networks: A telephone company laying down cables in a new neighborhood while making sure every house has a connection



Forest Management: Building pathways or roads to access different parts of a forest (for conservation, recreation, or management purposes) while minimizing environmental impact



Electrical Circuit Design: Electrical engineers designing circuits, and ensuring connectivity between various components while using the least amount of wiring

Minimum Spanning Tree Formulation

Let x_{ij} be 1 if edge ij is in the tree T while there are:

- $n - 1$ edges in T
- no cycles in T

$$\text{minimize } \sum_{ij \in E} c_{ij} x_{ij}$$

$$\text{subject to } \sum_{ij \in E} x_{ij} = n - 1$$

$$\sum_{ij \in E: i \in S, j \in S} x_{ij} \leq |S| - 1 \quad \forall S \subseteq V$$

$$x_{ij} \in \{0, 1\} \quad \forall ij \in E$$

LP Relaxation

LP Relaxation

$$\begin{aligned} &\text{minimize } \sum_{ij \in E} c_{ij} x_{ij} \\ &\text{subject to } \sum_{ij \in E} x_{ij} = n - 1 \\ &\quad \sum_{ij \in E: i \in S, j \in S} x_{ij} \leq |S| - 1 \quad \forall S \subseteq V \\ &\quad x_{ij} \in \{0, 1\} \quad \forall ij \in E \end{aligned}$$

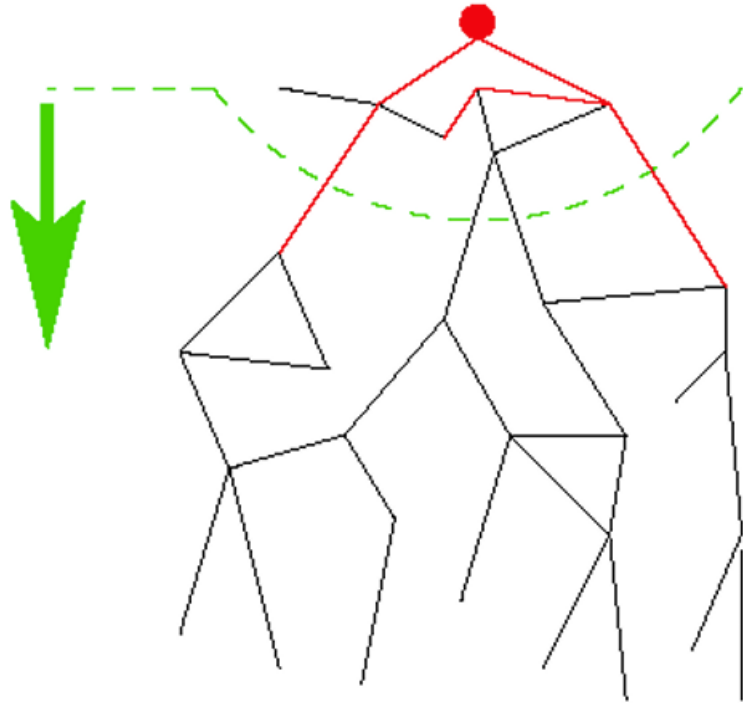
$$\begin{aligned} &\text{minimize } \sum_{ij \in E} c_{ij} x_{ij} \\ &\text{subject to } \sum_{ij \in E} x_{ij} = n - 1 \\ &\quad \sum_{ij \in E: i \in S, j \in S} x_{ij} \leq |S| - 1 \quad \forall S \subseteq V \\ &\quad x_{ij} \geq 0 \quad \forall ij \in E \end{aligned}$$



MST Algorithms

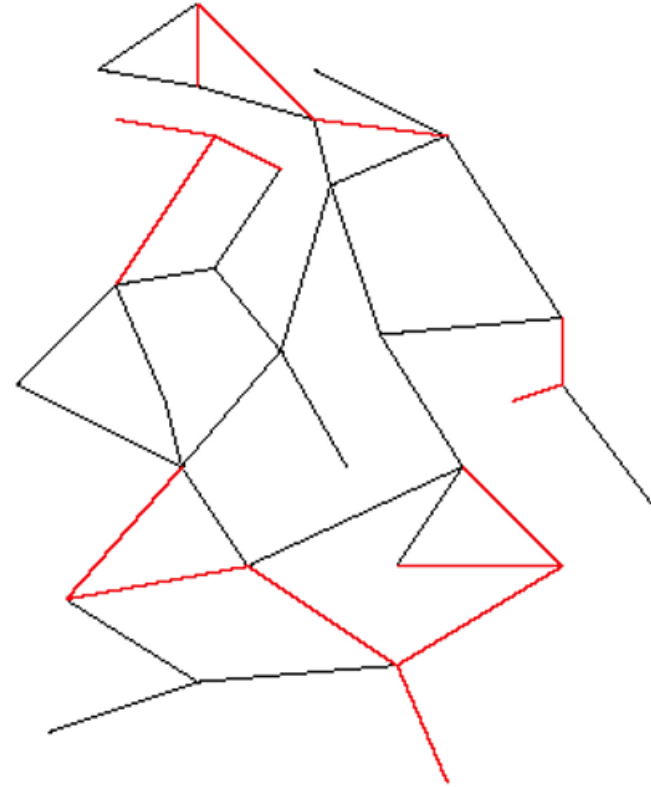
- **Prim's Algorithm**
- **Kruskal's Algorithm**

MST Algorithms: Two Different Approaches



Prim's Algorithm

Start with one node, grow greedily



Kruskal's Algorithm

Start with a forest of MSTs, union them together

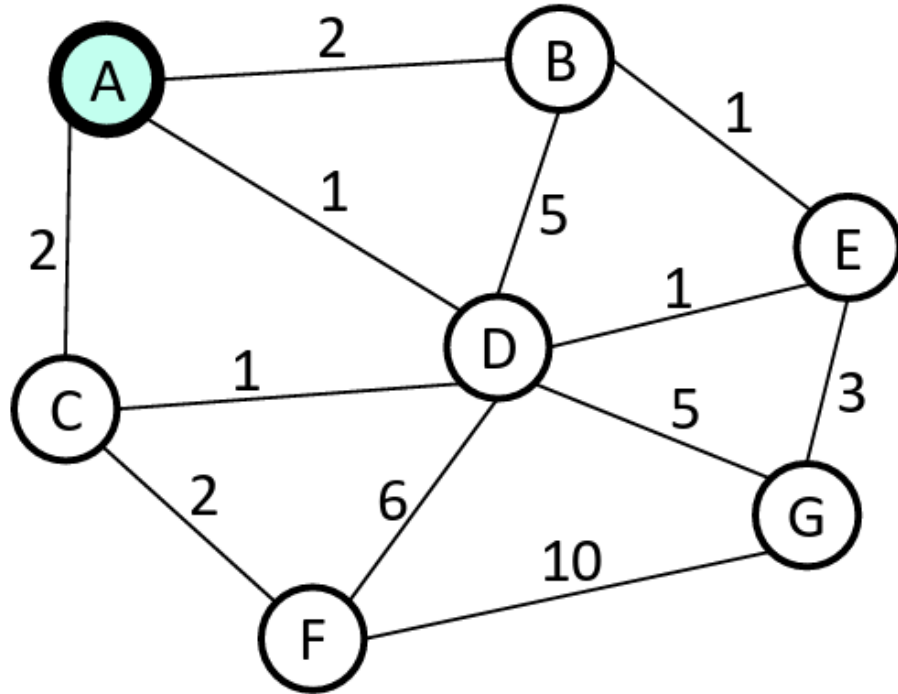
Prim's Algorithm

- This is a node-based greedy algorithm
 - Builds MST by greedily adding nodes/vertices
- Grows a single tree by picking a node from the fringe* that has the smallest cost/weight

What algorithm does this remind you of?

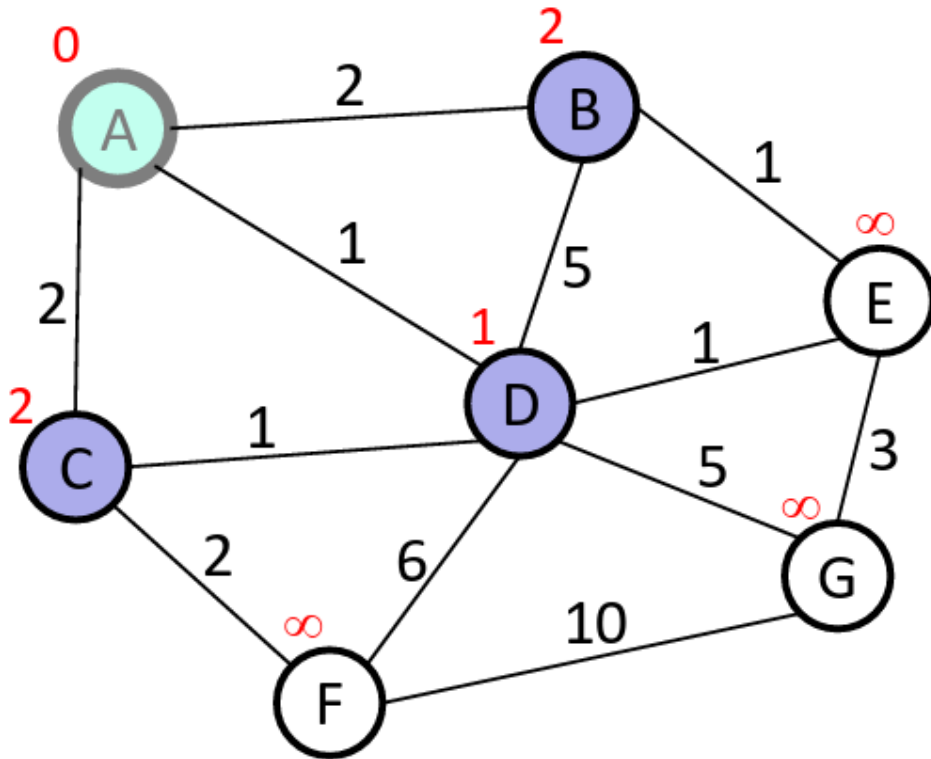
*A fringe is the set of nodes outside the MST that have at least one edge connecting them to the current MST

Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A		∞	
B		∞	
C		∞	
D		∞	
E		∞	
F		∞	
G		∞	

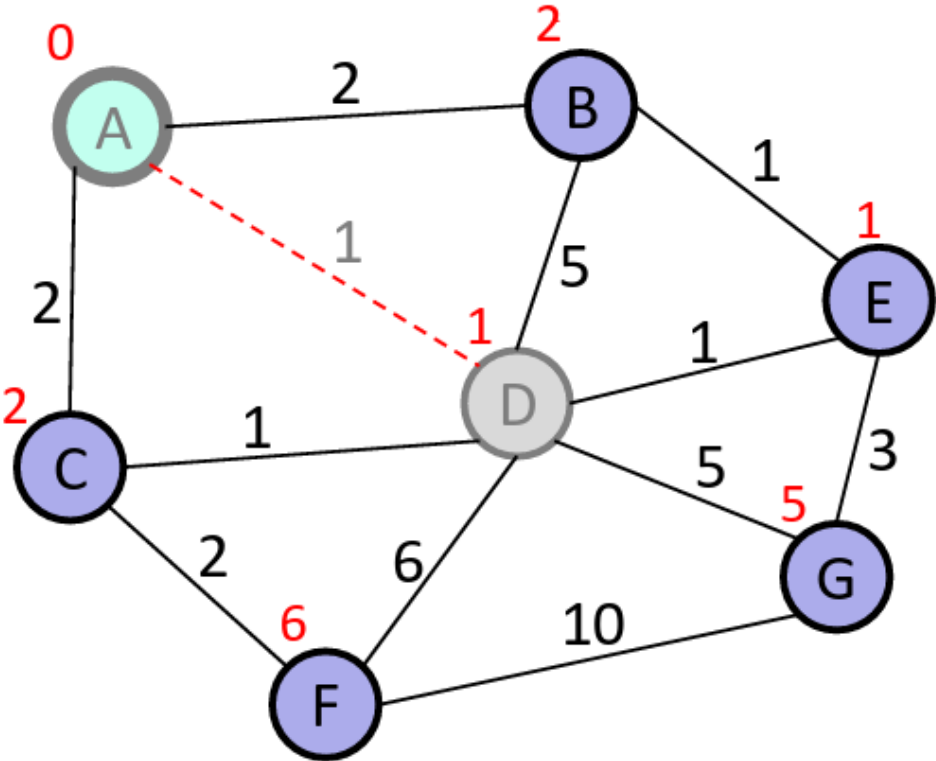
Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B		2	A
C		2	A
D		1	A
E		∞	
F		∞	
G		∞	

Added to set: A

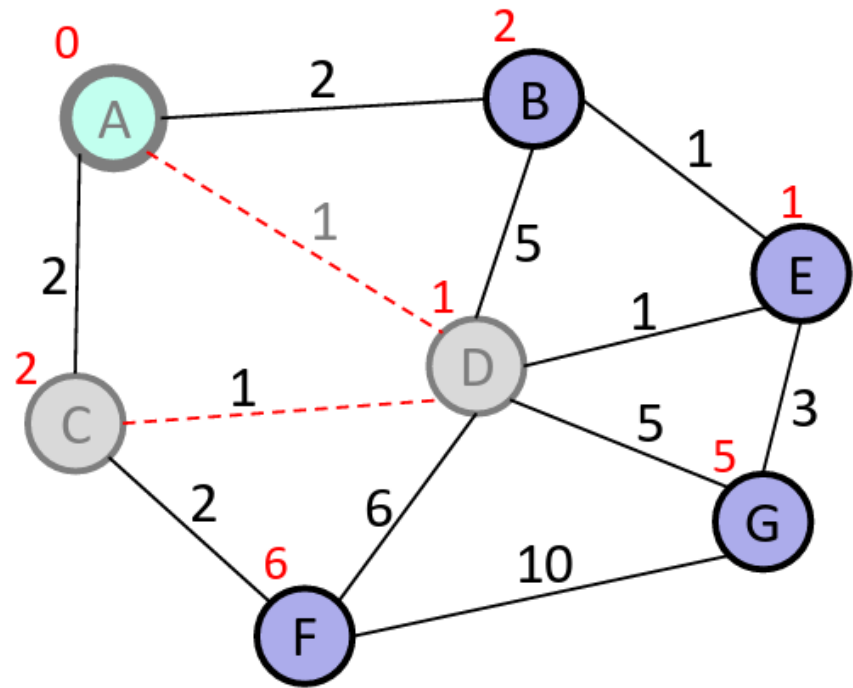
Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B		2	A
C		1	D
D	Y	1	A
E		1	D
F		6	D
G		5	D

Added to set: A,D

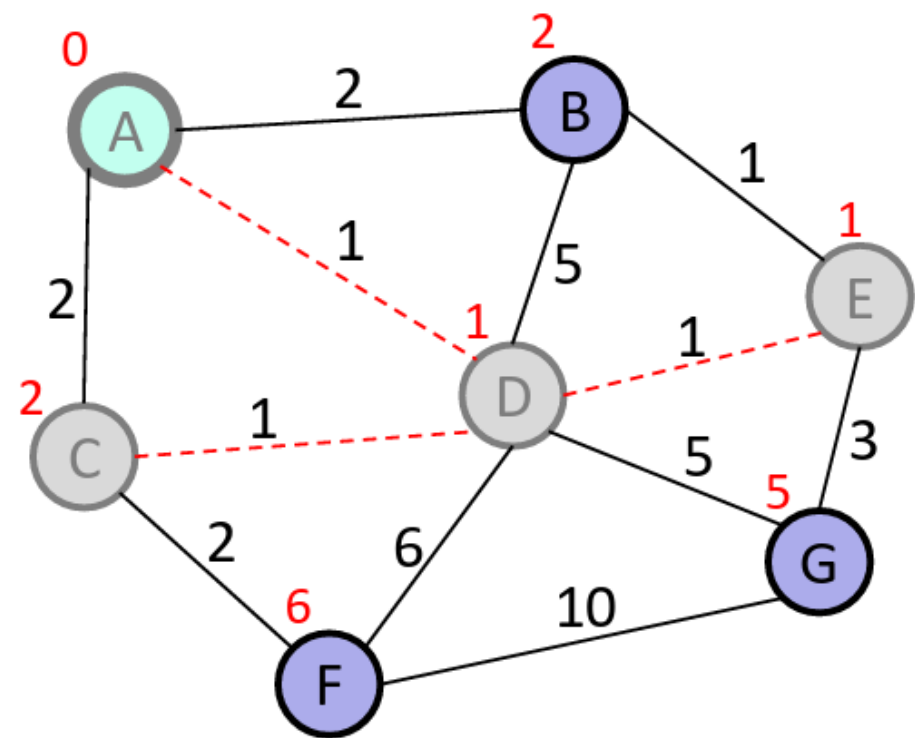
Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B		2	A
C	Y	1	D
D	Y	1	A
E		1	D
F		2	C
G		5	D

Added to set: A,D,C

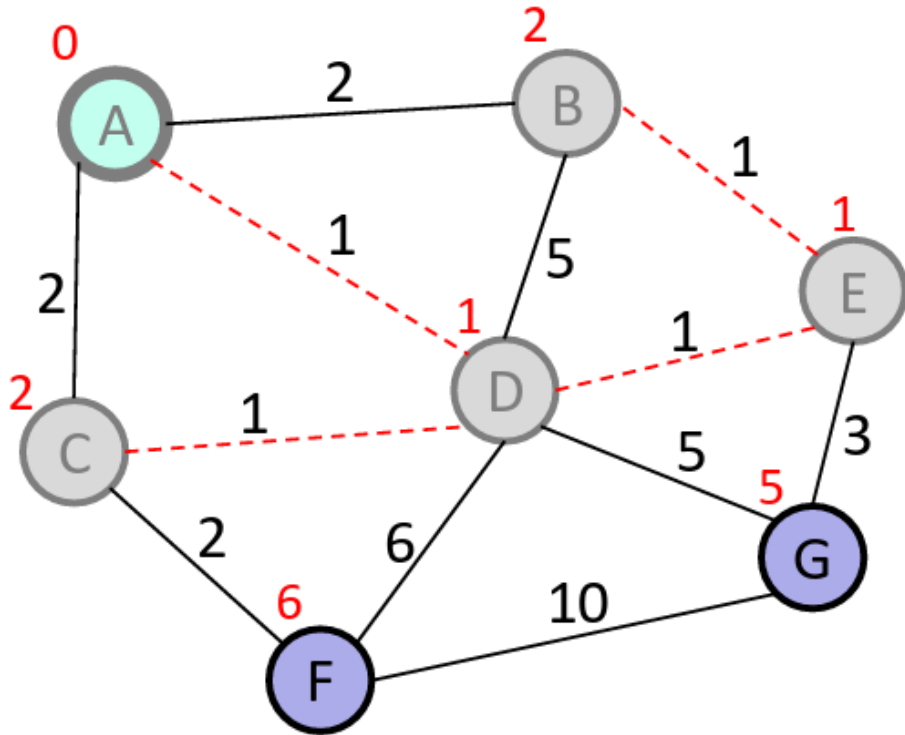
Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B		1	E
C	Y	1	D
D	Y	1	A
E	Y	1	D
F		2	C
G		3	E

Added to set: A,D,C,E

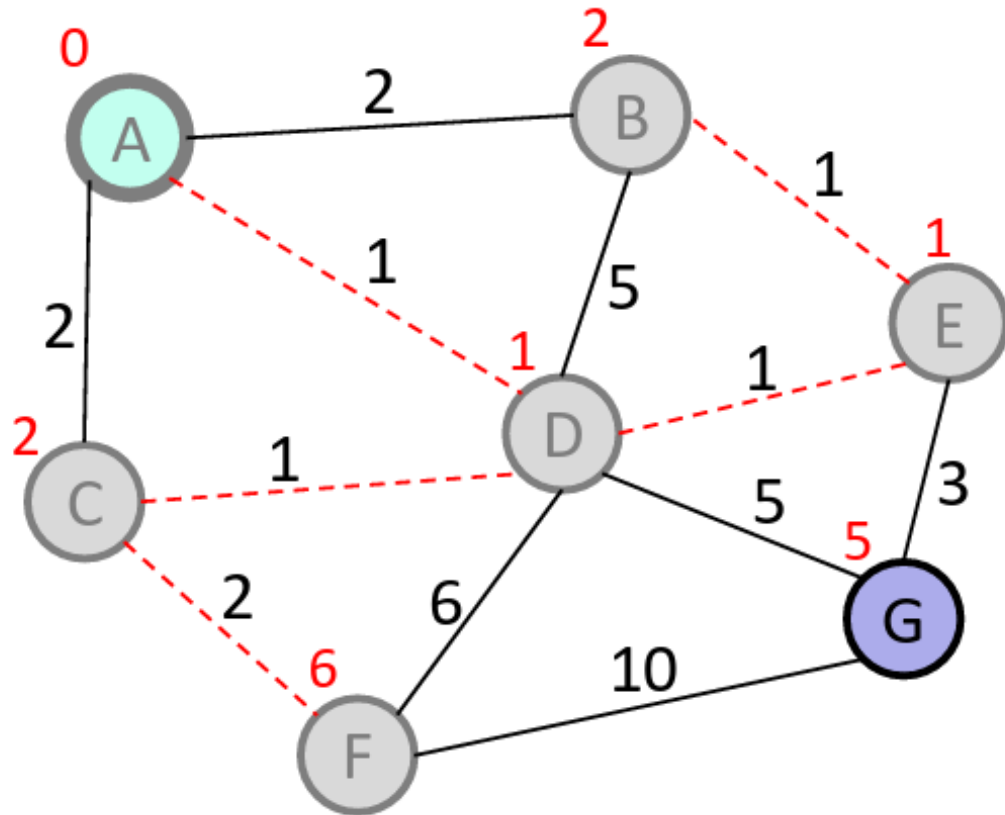
Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B	Y	1	E
C	Y	1	D
D	Y	1	A
E	Y	1	D
F		2	C
G		3	E

Added to set: A,D,C,E,B

Prim's Algorithm | Example



Vertex	Known?	Distance	Previous
A	Y	0	-
B	Y	1	E
C	Y	1	D
D	Y	1	A
E	Y	1	D
F	Y	2	C
G		3	E

Added to set: A,D,C,E,B,F

Kruskal's Algorithm: A Different Approach

- Prim's thinks node by node
 - Adds the closest node to the currently reachable set
- What if you think edge by edge instead?
 - Eg, start from the lightest edge; add it if it connects new nodes to each other (and don't add it if it would create a cycle)
- This is an edge-based greedy algorithm
 - Builds MST by greedily adding edges

Kruskal's Algorithm in a Nutshell



Step 1: Define T to be empty, add edges are unprocessed

Kruskal's Algorithm



Step 1: Define T to be empty, add edges are unprocessed

Step 2: Find an unprocessed edge e of smallest weight w_e

Kruskal's Algorithm



Step 1: Define T to be empty, add edges are unprocessed

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Step 3: If $(V, T \cup \{e\})$ is a forest, then add e to T

Kruskal's Algorithm



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Step 4: Mark e as processed and

Kruskal's Algorithm



Step 1: Define T to be empty, add edges are unprocessed

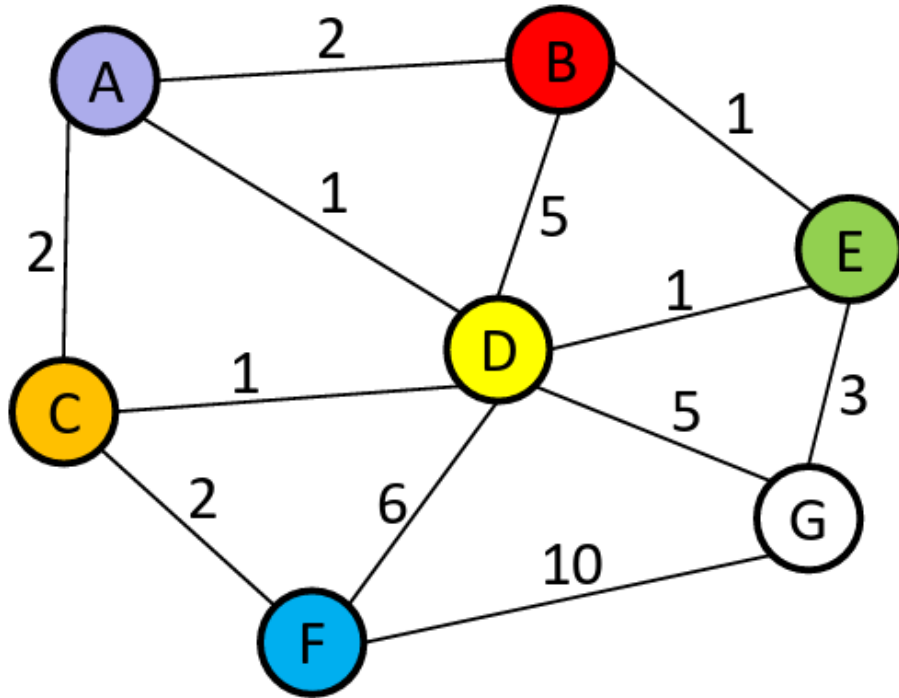
Step 2: Find an unprocessed edge e of smallest weight w_e

Step 3: If $(V, T \cup \{e\})$ is a forest, then add e to T

Step 4: Mark e as processed and

Step 5: Repeat until all edges are processed

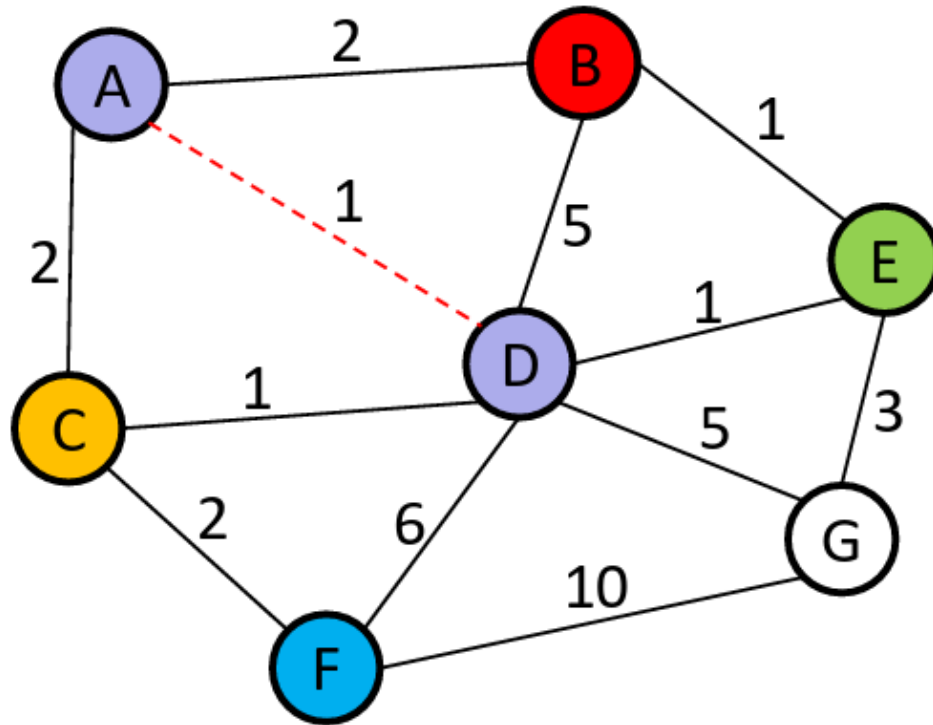
Kruskal's Algorithm | Example



Weight	Edges
1	(A,D), (C,D), (B,E), (D,E)
2	(A,B), (C,F), (A,C)
3	(E,G)
5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST:

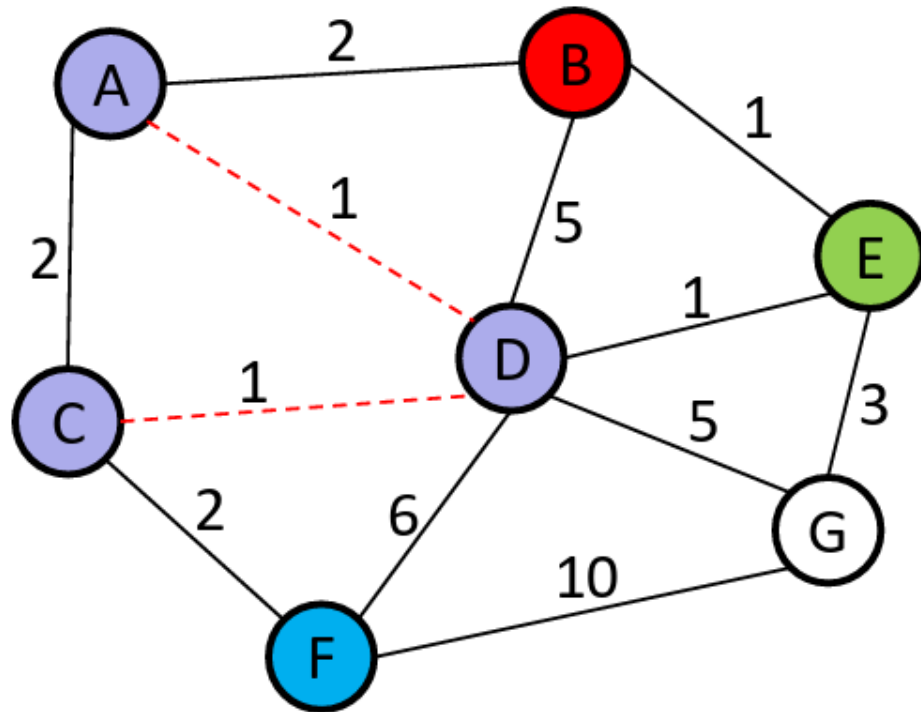
Kruskal's Algorithm | Example



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3	(E,G)
5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST: (A,D)

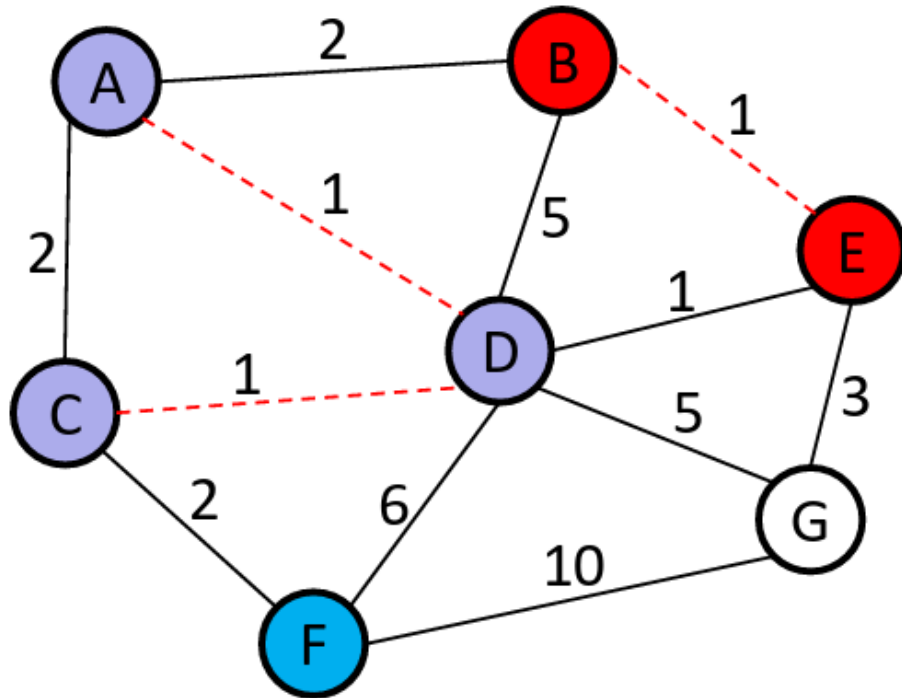
Kruskal's Algorithm | Example



Weight	Edges
1	(A,D), (C,D), (B,E), (D,E)
2	(A,B), (C,F), (A,C)
3	(E,G)
5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST: (A,D), (C,D)

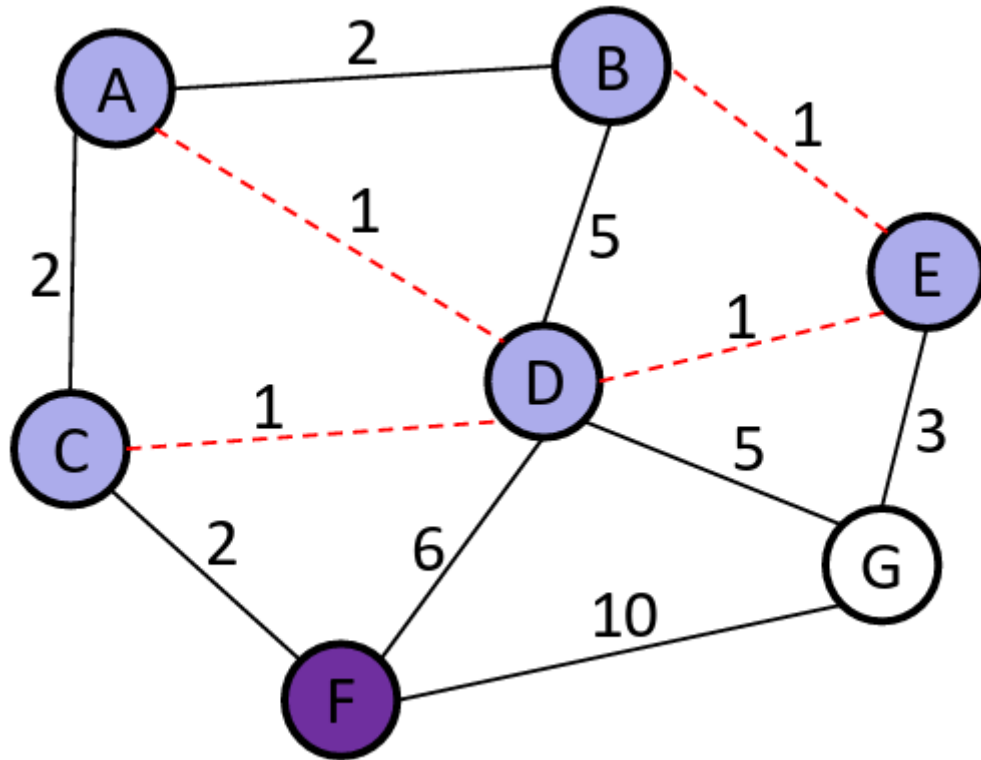
Kruskal's Algorithm | Example



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5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST: (A,D), (C,D), (B,E)

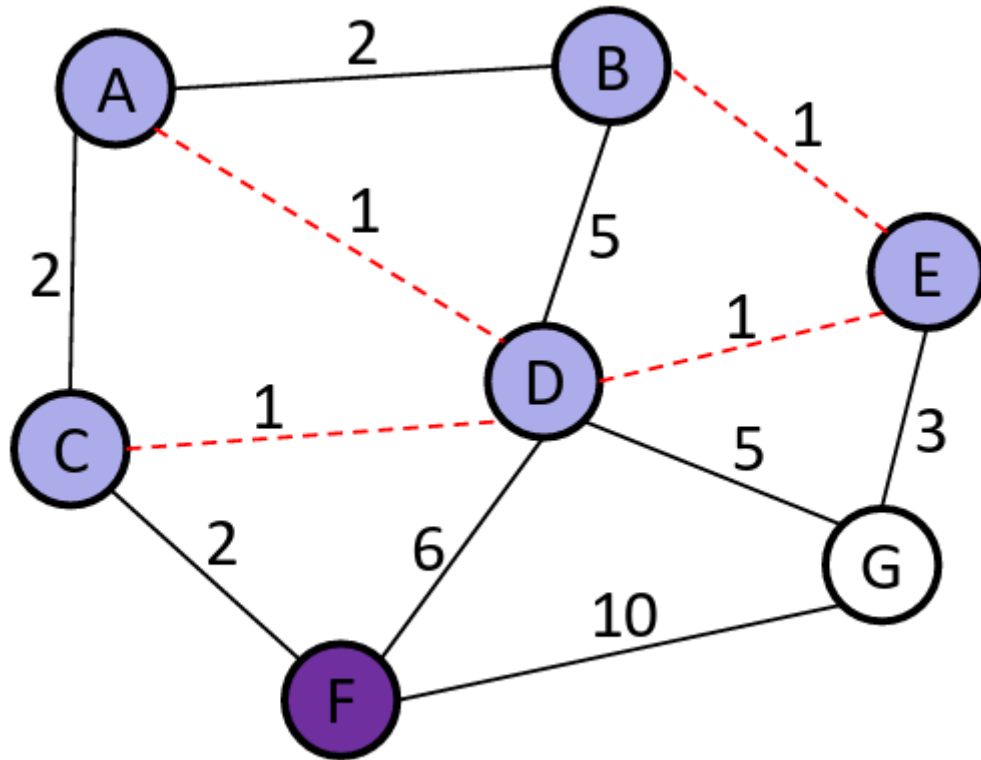
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MST: (A,D), (C,D), (B,E), (D,E)

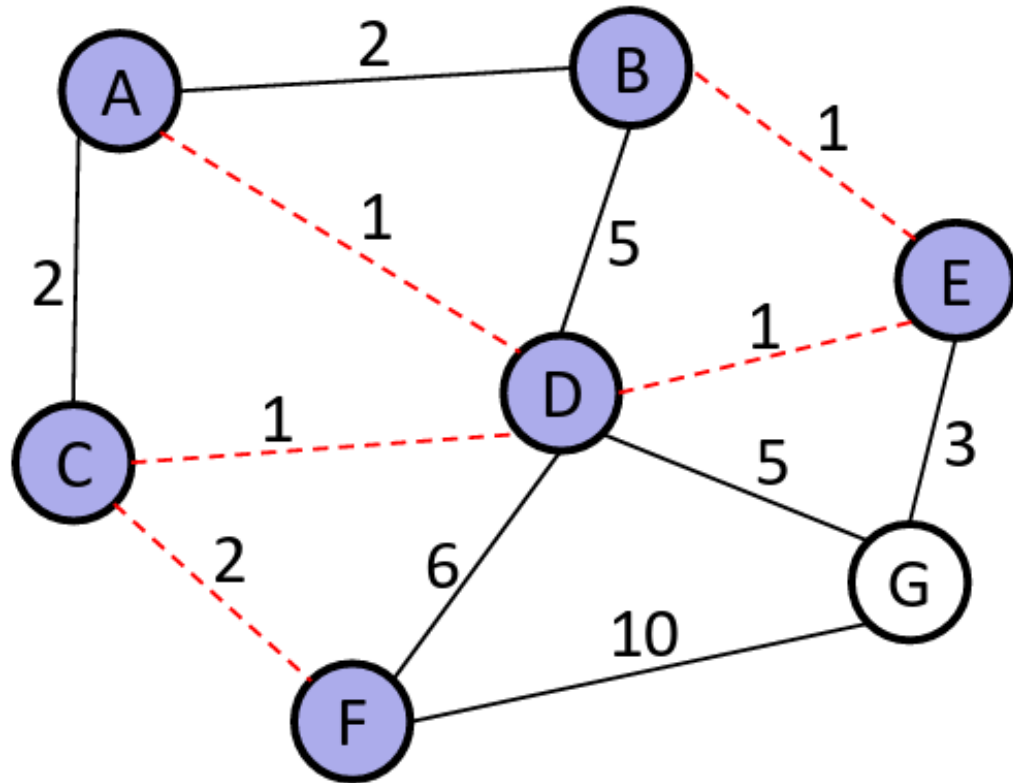
Kruskal's Algorithm | Example



Weight	Edges
1	(A,D), (C,D), (B,E), (D,E)
2	(A,B), (C,F), (A,C)
3	(E,G)
5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST: (A,D), (C,D), (B,E), (D,E)

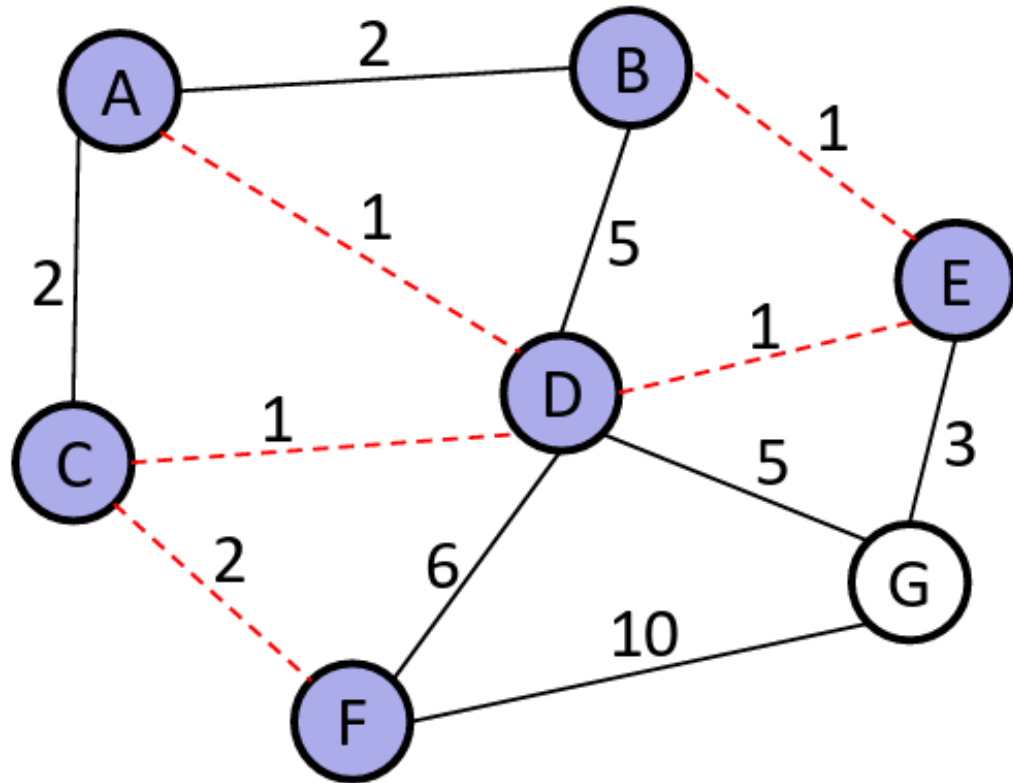
Kruskal's Algorithm | Example



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6	(D,F)
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MST: (A,D), (C,D), (B,E), (D,E), (C,F)

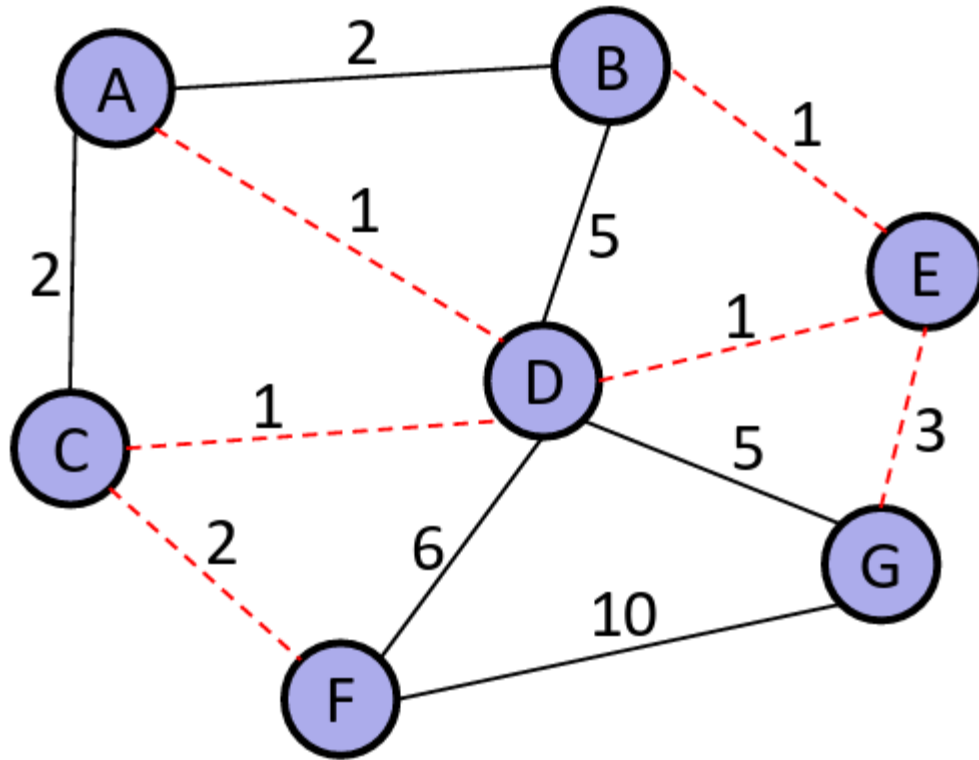
Kruskal's Algorithm | Example



Weight	Edges
1	(A,D), (C,D), (B,E), (D,E)
2	(A,B), (C,F), (A,C)
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6	(D,F)
10	(F,G)

MST: (A,D), (C,D), (B,E), (D,E), (C,F)

Kruskal's Algorithm | Example



Weight	Edges
1	(A,D), (C,D), (B,E), (D,E)
2	(A,B), (C,F), (A,C)
3	(E,G)
5	(D,G), (B,D)
6	(D,F)
10	(F,G)

MST: (A,D), (C,D), (B,E), (D,E), (C,F), (E,G) $\longrightarrow$ **Total Length: 9**

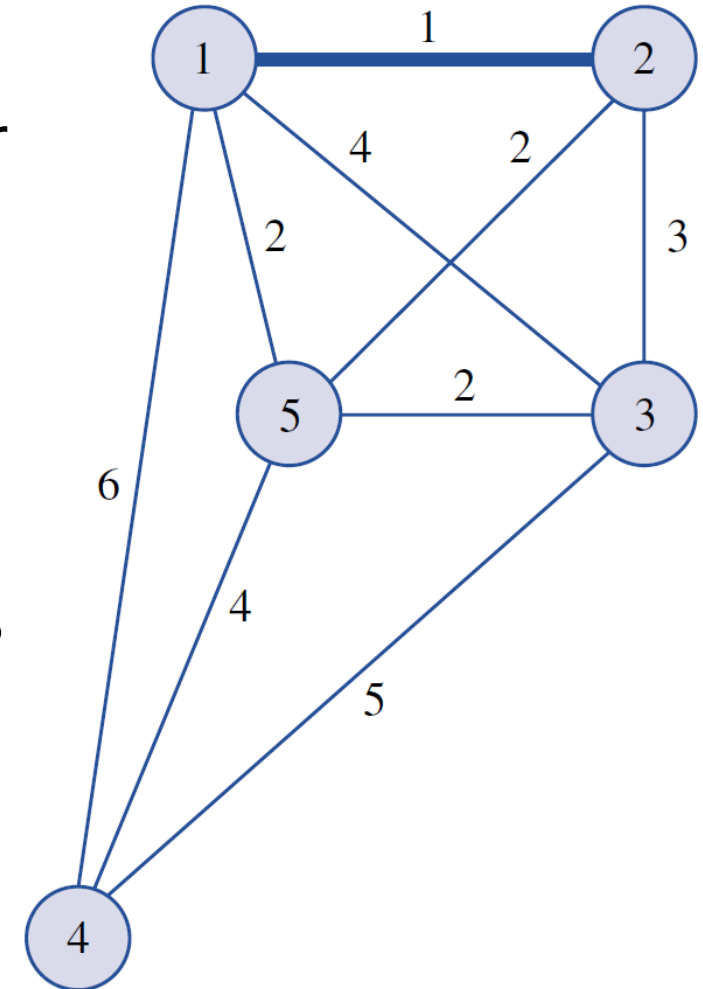
Example: Minimum Spanning Tree

The State University campus has five minicomputers. The distance between each pair of computers (in city blocks) is given in the network.

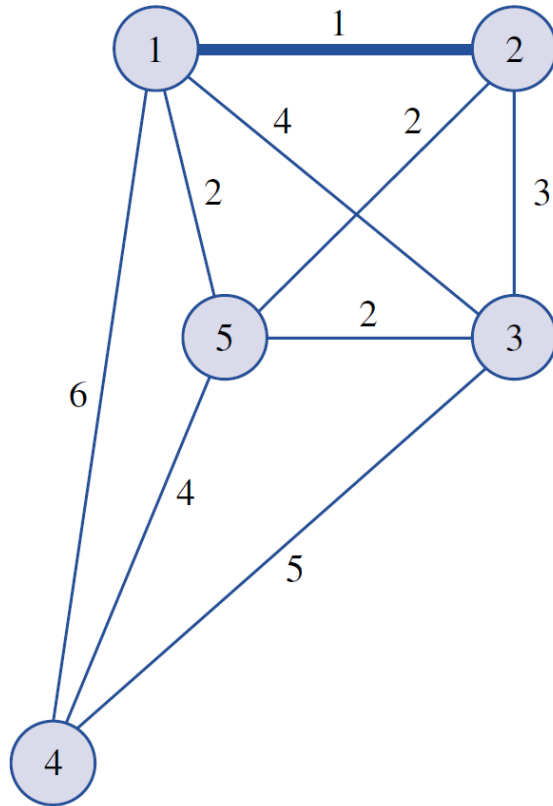
The computers must be interconnected by underground cable.

What is the minimum length of cable required?

Note that if no arc is drawn connecting a pair of nodes, this means that no cable can be laid between these two computers.

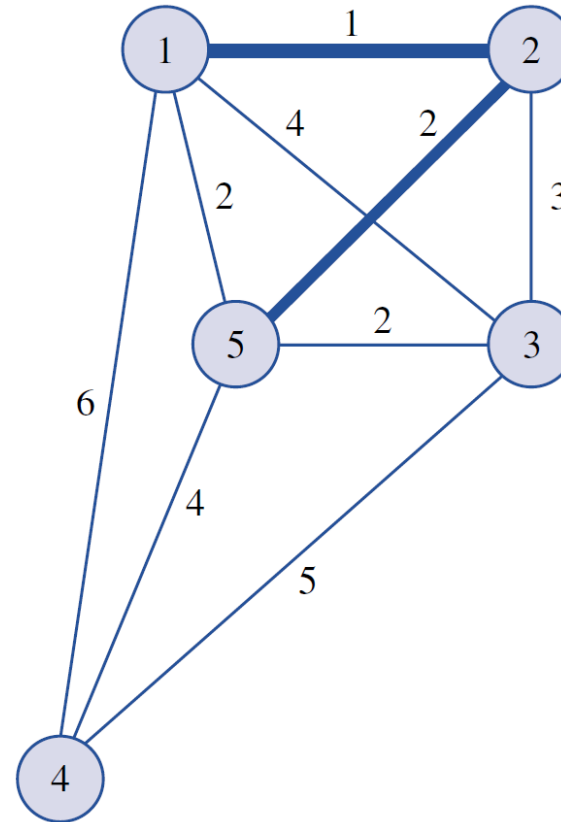


Example: Minimum Spanning Tree



$C = [1, 2]$
 $C' = [3, 4, 5]$

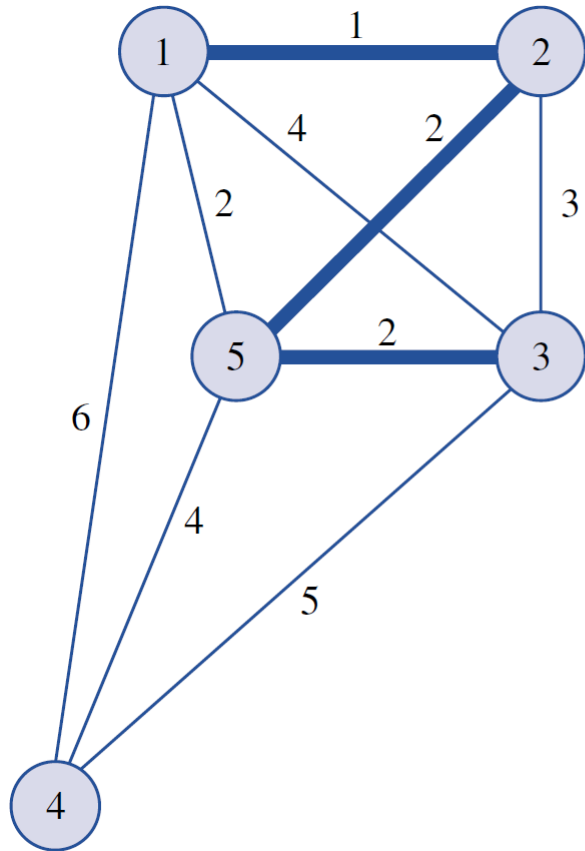
Iteration 1



$C = [1, 2, 5]$
 $C' = [3, 4]$

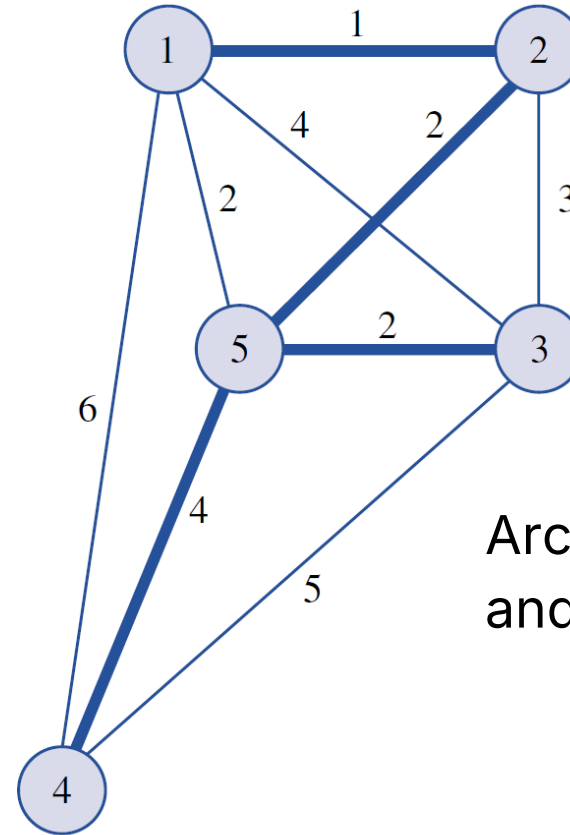
Iteration 2

Example: Minimum Spanning Tree



Iteration 3

$C = [1, 2, 3, 5]$
 $C' = [4]$



Arcs (1, 2), (2, 5), (5, 3),
and (5, 4) are the MST

Iteration 4

The Offshore Wind Farming Problem

